RAJALAKSHMI ENGINEERING COLLEGE

Sagana R 251001223-2025-IT S2

Started on	Friday, 14 August 2026, 1:53 PM
State	Finished
Completed on	Friday, 21 August 2026, 1:40 PM
Time taken	6 days 23 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n;
    scanf("%d", &n);

    int count = 0;
    int m = n / 2;
    int l = 0;

    scanf("%d", &n);

    m = n - n / 2;

    for (int x = 1; x < n; x = x * 2)
        l++;

    count = 1 + (m + 1) + (m * (l + 1)) + (m * l * (l + 1));

    /* c++ executions */
    count += m * l * l;

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔

	Input	Expected	Got	
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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